# Package 'golemworldbank'

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**Title** World Bank Indicators Explorer

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**Author** Dimitrios Zacharatos [aut, cre]

**Maintainer** Dimitrios Zacharatos <dzacharatos@yahoo.com>

**Description** A Shiny application, built as a golem package, for exploring World Bank development indicators through interactive country comparison, indicator comparison, population pyramid, ranked bar chart, correlation scatter, and choropleth map views.

**License** GPL-3

**URL** https://sedzinfo.github.io/golemworldbank/, https://github.com/sedzinfo/golemworldbank

**BugReports** https://github.com/sedzinfo/golemworldbank/issues

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`animation_buttons` *Pause and Stop buttons for an animated plotly chart*

---

## Description

Builds the `updatemenus` entry shown next to plotly's own Play button, used by every animated chart of the app. Pause halts the animation on the current frame; Stop halts it and jumps back to the first frame.

## Usage

```
animation_buttons(first_frame)
```

## Arguments

| | |
|---|---|
| `first_frame` | Character. Name of the first animation frame (for these charts, the first year). |

## Value

A list to pass as `plotly::layout(updatemenus = ...)`.

## Examples

```
plotly::layout(p, updatemenus = animation_buttons("1960"))
```

## country_code *Country and region reference table*

### Description

The World Bank metadata of each country and aggregate: its names, region, income group and statistical practices. The app uses `Short Name`, matched to `Country Name` of mfi, and `Region` (empty for the aggregates), to color the bubbles of the Scatterplot tab by region.

### Usage

```
country_code
```

### Format

A data frame with 265 rows and 31 columns, including `Country Code`, `Short Name`, `Table Name`, `Long Name`, `Region`, and `Income Group`.

### Source

World Bank Open Data (https://data.worldbank.org).

### Examples

```
head(country_code)
```

## format_value *Format numbers compactly to 3 significant digits*

### Description

Abbreviates numbers to 3 significant digits with the suffixes k, M, B and T (thousands, millions, billions, trillions); negative and small values keep their sign and digits (6.02M, 1.42B, 0.0752, -5.72B). Used for the labels and the hover text of the bars in `plot_bar()`.

### Usage

```
format_value(n)
```

### Arguments

| | |
|---|---|
| `n` | Numeric vector of values to format. |

### Value

A character vector of formatted values.

## Examples

```
format_value(c(0.0752, 950, 1520, 6020000, -5720000000, 29200000000000))
#> [1] "0.0752" "950" "1.52k" "6.02M" "-5.72B" "29.2T"
```

---

`indicator_change` *Indicators that rose or fell between two years for one country*

---

## Description

Compares every indicator a country has a value for in both years, for the Change (%) tab, and splits them into two tables: the ones that increased from `year1` to `year2` and the ones that decreased, each sorted by percent change (largest change first). The percent change is `(value2 - value1) / |value1| * 100`; it is undefined when the `year1` value is 0, so those indicators are left out of both tables (their names are kept in the `zero_base` element). Indicators that did not change are in neither table.

Rates of change ("annual %", "% change", "growth", e.g. "GDP growth (annual %)", "Inflation, consumer prices (annual %)", "... (% change from 1990)") are left out by default: a percent change of a growth rate is not meaningful (a rate going from -0.01 to 1.7 reads as +17,000%). `include_rates = TRUE` keeps them.

For indicators that are themselves percentages (e.g. "Urban population (% of total population)") the percent change is relative to the old percentage, not a difference in percentage points; the `Change` column gives the plain difference.

## Usage

```
indicator_change(
  mfi,
  country,
  year1,
  year2,
  include_rates = FALSE,
  include_sign_change = FALSE
)
```

## Arguments

| | |
|---|---|
| `mfi` | Data frame of World Bank indicator values (long format, with `Country Name`, `Indicator Name`, `Year`, `value` columns). |
| `country` | Character. Country name, as in `Country Name`. |
| `year1`, `year2` | Numeric or character. The two years to compare. |
| `include_rates` | Logical. Keep the rate-of-change indicators (names containing "annual %", "% change" or "growth"); `FALSE` (default) leaves them out. |
| `include_sign_change` | Logical. Keep indicators whose value changed sign between the two years (e.g. net FDI inflows from -8 million to +6.7 billion): their percent change is not meaningful (+82,000%). `FALSE` (default) leaves them out; either way they are listed in `sign_change`. |

## Value

A list with `increased` and `decreased` (data frames with the columns `Indicator`, one named after each year with its value, `Change` and `Change %`), `zero_base` (indicators left out because their `year1` value is 0) and `sign_change` (indicators whose value changed sign; left out of the tables unless `include_sign_change = TRUE`). Stops with an error when the country has no data in either year.

## Examples

```
res <- indicator_change(mfi, country = "Greece", year1 = 2000, year2 = 2020)
head(res$increased)
head(res$decreased)
```

---

# `indicator_info` *Indicator topics and definitions*

---

## Description

World Bank topic and definition of every indicator in mfi, shown in the Index tab.

## Usage

```
indicator_info
```

## Format

A data frame with 1,497 rows and 3 columns:

**Indicator Name** Indicator name, as in mfi.

**Topic** World Bank topic, e.g. "Health: Population: Structure".

**Definition** Short definition of the indicator.

## Source

World Bank Open Data (https://data.worldbank.org), WDISeries.csv.

## Examples

```
head(indicator_info)
```

## keep_legend_clicks *Keep the legend clicks of an animated plotly chart*

### Description

plotly writes `visible` into every trace of every animation frame, so a trace hidden with a click on the legend comes back on the next frame. This removes `visible` from the frames, so the click lasts while the animation plays. plotly also hides a trace, with `visible = FALSE`, in a frame where it has no data, as a copy of its previous frame; that copy is emptied instead, so nothing is drawn there.

### Usage

```
keep_legend_clicks(p)
```

### Arguments

| | |
|---|---|
| p | A `plotly` object with frames. |

### Value

The built `plotly` object.

### Examples

```
keep_legend_clicks(plot_pyramid(mfi_population, country = "Greece", font_style = font_style))
```

## mfi *World Bank indicator values*

### Description

The values of the World Development Indicators in long format, one row per country, indicator and year: 217 countries, 1,497 indicators, the years 1960 to 2025. Regions and income groups (such as "World") are not included, and neither are missing values. This is the data of every tab except the Pyramid.

### Usage

```
mfi
```

## Format

A data frame with 7,505,749 rows and 5 columns:

**Country Code** Three-letter World Bank code of the country, e.g. "GRC".

**Country Name** Country name, e.g. "Greece".

**Indicator Name** Indicator name, e.g. "Population, total".

**Year** Year of observation (factor).

**value** Value of the indicator for that country and year.

## Source

World Bank Open Data (https://data.worldbank.org).

## Examples

```
head(mfi)
```

---

# mfi_population *Population by age group and sex*

---

## Description

The share of each five-year age group in the male and in the female population, per country and year: the 32 indicators of mfi such as "Population ages 40-44, male (% of male population)", with their age group and sex in columns of their own. This is the data of the Pyramid tab.

## Usage

```
mfi_population
```

## Format

A data frame with 458,304 rows and 7 columns:

**Country Name** Country name.

**Indicator Name** Indicator name, as in mfi.

**Year** Year of observation (factor).

**value** Percent of the male or of the female population.

**age** Age group (factor, "00-04" to "80+").

**unit** Unit of `value`: always "Percent".

**sex** "Male" or "Female".

## Source

World Bank Open Data (https://data.worldbank.org).

## Examples

```
head(mfi_population)
```

---

`plot_bar` *Build an animated ranked bar chart*

---

## Description

Plots the `top_n` countries with the largest value of one indicator as horizontal bars, animated over `Year` (a "bar chart race"), for the Barplot tab: in every frame the bars are ranked again, so a country that overtakes another moves up. A country's missing years between two observed ones are filled by linear interpolation, so that it does not drop out of the race for a year and come back; nothing is filled before its first or after its last observed year. Each bar carries the country name and its value, to 3 significant digits (1.52k, 6.02M); the bar of `country` is red, so that it can be followed through the race, and the year of the frame is printed large and gray in the bottom right corner. Only rows with a positive value are used.

## Usage

```
plot_bar(mfi, indicator, font_style, top_n = 20, country = NULL)
```

## Arguments

| | |
|---|---|
| `mfi` | Data frame of World Bank indicator values (long format, with `Country Name`, `Indicator Name`, `Year`, `value` columns). |
| `indicator` | Character. Indicator name to plot. |
| `font_style` | A `plotly` font list for the layout of the chart; the app passes font_style. |
| `top_n` | Integer. Number of countries shown in each year. |
| `country` | Character. Country whose bar is drawn red (`NULL` for none). |

## Value

A `plotly` htmlwidget, or `NULL` when the indicator has fewer than two positive values.

## Examples

```
plot_bar(mfi, indicator = "GDP (current US$)", font_style = font_style, country = "Greece", top_n = 50)
```

plot_cor *Build an animated correlation bubble scatter plot*

## Description

Plots every country as a bubble, at its value of `indicator1` (x) and of `indicator2` (y), sized by its population and colored by its region, animated over `Year`, for the Scatterplot tab. The region comes from `country_code`, by matching its `Short Name` to the `Country Name` of `mfi`; a country whose names differ there has no region and is left out, as is a country-year without both values and the population.

## Usage

```
plot_cor(mfi, country_code, indicator1, indicator2, font_style)
```

## Arguments

| | |
|---|---|
| `mfi` | Data frame of World Bank indicator values (long format, with `Country Name`, `Indicator Name`, `Year`, `value` columns). Must also contain a `"Population, total"` indicator, used for bubble sizing. |
| `country_code` | Data frame mapping countries to region, with columns `Country Code`, `Short Name`, `Region`. |
| `indicator1` | Character. Indicator name plotted on the x-axis. |
| `indicator2` | Character. Indicator name plotted on the y-axis. |
| `font_style` | A `plotly` font list for the layout of the chart; the app passes font_style. |

## Value

A `plotly` htmlwidget.

## Examples

```
plot_cor(
  mfi,
  country_code,
  indicator1 = "Mortality rate, adult, male (per 1,000 male adults)",
  indicator2 = "Mortality rate, infant (per 1,000 live births)",
  font_style = font_style
)
```

## plot_country_comp *Build a country comparison line chart*

### Description

Plots one indicator over the years, one line per country, for the Country Comparison tab. The hover text gives the country, indicator, year and value. Rows with a missing value are left out.

### Usage

```
plot_country_comp(mfi, countries, indicator, font_style)
```

### Arguments

| | |
|---|---|
| mfi | Data frame of World Bank indicator values (long format, with `Country Name`, `Indicator Name`, `Year`, `value` columns). |
| countries | Character vector of country names to plot. |
| indicator | Character. Indicator name to plot. |
| font_style | A `plotly` font list for the layout of the chart; the app passes font_style. |

### Value

A `plotly` htmlwidget.

### Examples

```
plot_country_comp(
  mfi,
  countries = c("Greece", "Bulgaria"),
  indicator = "Population, total",
  font_style = font_style
)
```

## plot_indicator_comp *Build an indicator comparison line chart*

### Description

Plots several indicators over the years for one country, one line per indicator, for the Indicator Comparison tab. The indicators share one y axis, so they compare well only when they have the same unit and a similar scale (e.g. the female and the male population). Rows with a missing value are left out.

### Usage

```
plot_indicator_comp(mfi, country, indicators, font_style)
```

### Arguments

| | |
|---|---|
| mfi | Data frame of World Bank indicator values (long format, with Country Name, Indicator Name, Year, value columns). |
| country | Character. Country name to plot. |
| indicators | Character vector of indicator names to plot. |
| font_style | A plotly font list for the layout of the chart; the app passes font_style. |

### Value

A plotly htmlwidget.

### Examples

```
plot_indicator_comp(
  mfi,
  country = "Greece",
  indicators = c("Population, female", "Population, male"),
  font_style = font_style
)
```

---

## plot_map *Build an animated choropleth map*

---

### Description

Colors every country of a world map by the value of one indicator, from white (lowest) to dark blue (highest), animated over Year, for the Map tab. Countries are placed by their Country Code. Rows with a missing value are left out.

### Usage

```
plot_map(mfi, indicator, font_style)
```

### Arguments

| | |
|---|---|
| mfi | Data frame of World Bank indicator values (long format, with Country Name, Country Code, Indicator Name, Year, value columns). |
| indicator | Character. Indicator name to plot. |
| font_style | A plotly font list for the layout of the chart; the app passes font_style. |

### Value

A plotly htmlwidget.

### Examples

```
plot_map(mfi, indicator = "Population, total", font_style = font_style)
```

## plot_pyramid *Build an animated population pyramid*

### Description

Plots, for one country, the share of each five-year age group in the male population (bars to the left) and in the female population (to the right), animated over `Year`, for the Pyramid tab. The male values are negated so that the two sexes stand back to back; the labels on the bars show them positive, and so do the axis ticks (-5 reads "5%").

### Usage

```
plot_pyramid(mfi_population, country, font_style)
```

### Arguments

| | |
|---|---|
| `mfi_population` | Data frame of population shares by age group and sex (long format, with `Country Name`, `sex`, `age`, `Year`, `value` columns), see mfi_population. |
| `country` | Character. Country name to plot. |
| `font_style` | A `plotly` font list for the layout of the chart; the app passes font_style. |

### Value

A `plotly` htmlwidget.

### Examples

```
plot_pyramid(mfi_population, country = "Greece", font_style = font_style)
```

## render_change_table *Render one table of* `indicator_change()` *as a DataTable*

### Description

The values of the two years and the change are shown to 3 significant digits (6.02M, 1.42B, 0.0752), while sorting still uses the full numbers. The percent change has one decimal, in bold green for an increase and red for a decrease. The table is one long page with compact rows, like the Index table.

### Usage

```
render_change_table(tab)
```

### Arguments

| | |
|---|---|
| `tab` | One of the data frames returned by `indicator_change()` (`increased` or `decreased`). |

### Value

A `DT::datatable` htmlwidget.

### Examples

```
res <- indicator_change(mfi, country = "Greece", year1 = 2000, year2 = 2020)
render_change_table(res$increased)
```

---

## render_index_table *Build the indicator index DataTable*

---

### Description

Lists every indicator of `mfi`, in the order it first appears there, with its World Bank topic and definition, for the Index tab. The table is one long page, which can be searched and sorted.

### Usage

```
render_index_table(mfi, indicator_info)
```

### Arguments

| | |
|---|---|
| `mfi` | Data frame of World Bank indicator values, with an `Indicator Name` column. |
| `indicator_info` | Data frame with `Indicator Name`, `Topic` and `Definition` columns (see indicator_info). |

### Value

A `DT::datatable` htmlwidget.

### Examples

```
render_index_table(mfi, indicator_info)
```

---

## run_app *Run the World Bank Indicators Explorer*

---

### Description

Builds the app: country and indicator comparisons over time, a population pyramid, a bar chart race, a bubble scatter plot, a world map, the indicators that changed between two years, and an index of the indicators with their definitions. The app has a dark mode, switched by the button at the right of the navbar.

It also sets `options(scipen = 999)`, so that R does not print numbers in scientific notation; the option stays set after the app stops.

## Usage

```
run_app(
  analytics = FALSE,
  onStart = NULL,
  options = list(),
  enableBookmarking = NULL,
  uiPattern = "/",
  ...
)
```

## Arguments

| | |
|---|---|
| analytics | TRUE to add Google Analytics, through the cookie banner shared with the sites of sedzinfo.github.io, which loads it only after the visitor accepts. Meant for the public deployment (see app.R); off by default, so that running the app yourself sends nothing. |
| onStart | See shiny::shinyApp(). |
| options | See shiny::shinyApp(). |
| enableBookmarking | See shiny::shinyApp(). |
| uiPattern | See shiny::shinyApp(). |
| ... | Options for the app, which its code reads with golem::get_golem_options(); see golem::with_golem_options(). |

## Value

A Shiny app object. Printing it, as the console does, starts the app.

## Examples

```
## Not run:
run_app()

# override any shiny::shinyApp() option, e.g. to pick a fixed port:
run_app(options = list(port = 8080))

## End(Not run)
```

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